Western MA Per Pupil Expenditure Report

With selected districts for comparison

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Appendices

Appendix A. Data Sources & Calculation Methodology

1. Data Sources

District Expenditures by Spending Category: Source: Massachusetts Department of Elementary and Secondary Education (DESE) Last updated: August 12, 2025 URL: https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/

This dataset provides detailed expenditure data by category for all Massachusetts school districts, including: • Expenditures per pupil (EPP) by spending category • Student enrollment (In-District FTE, Out-of-District FTE, Total FTE) • Annual data from FY1993 to present

Chapter 70 District Profiles: Source: Massachusetts Department of Elementary and Secondary Education (DESE) URL: Various sources compiled in profile_DataC70 spreadsheet tab

From the Ch70 website: "Chapter 70 District Profiles: The on-line Chapter 70 database shows, for each school district, yearly spending and state aid totals in comparison to the foundation budget. Trend data is available for each year going back to FY1993."

This dataset provides: • Chapter 70 Aid (c70aid): State aid received by each district • Required Net School Spending (rqdnss2): Minimum spending required by state law • Actual Net School Spending (actualNSS): Total district spending on education • Foundation Enrollment (distfoundenro): District foundation enrollment used for Ch70 funding calculations

Regional Classifications: Source: DESE district profiles and regional service mappings • Districts classified as Western MA based on EOHHS regional designations • School type classifications (Traditional, Regional, Charter, etc.)

2. Per-Pupil Expenditure (PPE) Definition

Per-pupil expenditure (PPE) is reported in the End of Year Report (EOYR) for municipal and regional districts, and is calculated by in-district FTE.

Per the DESE's Researcher's Guide, section XV. Using financial data: *"The out-of-district total cannot be properly reported as a per-pupil expenditure because the cost of tuitions varies greatly depending on the reason for going out of district."*

3. Compound Annual Growth Rate (CAGR)

For a budget that grows at variable rates year-over-year, CAGR tells us: "If growth had been perfectly steady over this period, what annual rate would produce the same start-to-end result?" It's the geometric mean growth rate, smoothing out year-to-year volatility into a single comparable number.

Formula: CAGR = (End_Value / Start_Value)^(1 / Years) – 1

Example: If expenditures grow from $10,000 to $12,000 over 5 years: CAGR = ($12,000 / $10,000)^(1/5) – 1 = 1.2^0.2 – 1 = 0.0371 = 3.71% per year

Note: CAGR requires positive values at both endpoints and is undefined if data is missing.

3a. Year-over-Year (YoY) Growth Rate

YoY growth rate measures the percentage change from one year to the next, providing insight into annual fluctuations.

Formula: YoY Growth = (Value_year / Value_previous_year – 1) × 100

Example: If expenditures increase from $10,000 (Year 1) to $10,500 (Year 2): YoY Growth = ($10,500 / $10,000 – 1) × 100 = 5.0%

If expenditures then increase to $11,000 (Year 3): YoY Growth = ($11,000 / $10,500 – 1) × 100 = 4.76%

Key differences from CAGR: • YoY shows year-to-year changes; CAGR smooths growth over multiple years • YoY captures annual volatility; CAGR assumes constant growth • YoY is useful for identifying specific years with unusual growth patterns

4. Enrollment-Based Cohorts

Districts are grouped by total in-district FTE enrollment into five cohorts (based on IQR analysis) for meaningful peer comparison:

Tiny (0-200 FTE): 17 districts (Cohort 1: below Q1) **Member districts:** conway, erving, farmington river reg, florida, hancock, hawlemont, leverett, new salem-wendell, pelham, petersham, richmond, rowe, savoy, shutesbury, sunderland, and 2 others

Small (201-800 FTE): 13 districts (Cohort 2: Q1 to median) **Member districts:** clarksburg, deerfield, frontier, gateway, granby, hadley, hatfield, lee, lenox, orange, pioneer valley, ralph c mahar, southern berkshire

Medium (801-1600 FTE): 15 districts (Cohort 3: median to Q3) **Member districts:** amherst, amherst-pelham, berkshire hills, central berkshire, easthampton, gill-montague, greenfield, hoosac valley regional, mohawk trail, monson, mount greylock, north adams, palmer, southwick-tolland-granville regional school district, ware

Large (1601-10K FTE): 14 districts (Cohort 4: above Q3) **Member districts:** agawam, athol-royalston, belchertown, chicopee, east longmeadow, hampden-wilbraham, holyoke, longmeadow, ludlow, northampton, pittsfield, south hadley, west springfield, westfield

Outliers (Springfield >10K FTE): 1 district (Cohort 5: statistical outlier, analyzed separately) **Member districts:** springfield

Weighted EPP Formula (for aggregate calculations): For each category and year: Weighted_EPP = Σ(District_EPP × District_In-District_FTE) / Σ(District_In-District_FTE)

Example: District A spends $5,000/pupil (500 students), District B spends $6,000/pupil (300 students) Weighted average = ($5,000×500 + $6,000×300) / (500+300) = $5,375/pupil

Enrollment Calculation: For each series (In-District FTE, Out-of-District FTE) and year: Sum across all member districts in the enrollment group.

Appendix B. Calculations and Examples

Complete calculations and worked examples for all figures and tables

Purpose

This appendix provides calculations and worked examples for all figures and tables in the report. It combines step-by-step calculation procedures with detailed examples for verification.

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Figure 1: Year-over-Year (YoY) Growth Rates

Location: Executive Summary page

Source: executive_summary_plots.py, lines 44-63

Formula:

YoY Growth(year) = [(PPE(year) / PPE(year-1)) - 1] × 100

Calculation Steps:

1. For each district, get total PPE by year:

- Load expenditure data from df (all expense categories)

- Pivot to get PPE by category and year

- Sum across all categories: total_ppe[year] = sum(PPE_category[year] for all categories)

2. Calculate YoY growth:

- For year in [2010, 2011, ..., 2024]:

YoY[year] = [(total_ppe[year] / total_ppe[year-1]) - 1] × 100

- First year (2009) has no YoY value (no prior year)

3. For cohort aggregates:

- Get all districts in cohort

- Calculate weighted aggregate PPE (see Calculation #5)

- Apply same YoY formula to aggregate PPE series

Example (Amherst 2023-2024):

- PPE(2023) = $25,432.18

- PPE(2024) = $26,789.45

- YoY Growth = [(26,789.45 / 25,432.18) - 1] × 100 = 5.34%

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Figure 2: 5-Year CAGR (Compound Annual Growth Rate) - Grouped Bars

Location: Executive Summary page

Source: executive_summary_plots.py, lines 66-95

Formula:

CAGR = [(End_Value / Start_Value)^(1/Years) - 1] × 100

Calculation Steps:

1. Define three 5-year periods:

- Period 1: 2009-2014 (5 years)

- Period 2: 2014-2019 (5 years)

- Period 3: 2019-2024 (5 years)

2. For each district and each period:

- Get total_ppe[start_year] and total_ppe[end_year]

- CAGR = [(total_ppe[end] / total_ppe[start])^(1/5) - 1] × 100

3. For cohort aggregates:

- Use weighted aggregate PPE (see Calculation #5)

- Apply same CAGR formula

Example (Medium Cohort 2019-2024):

- Weighted PPE(2019) = $22,145.67

- Weighted PPE(2024) = $28,934.21

- CAGR = [(28,934.21 / 22,145.67)^(1/5) - 1] × 100

- CAGR = [1.30672^0.2 - 1] × 100

- CAGR = [1.05497 - 1] × 100 = 5.50%

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Figure 3: 15-Year CAGR (2009-2024)

Location: Executive Summary page

Source: executive_summary_plots.py, lines 98-118

Formula:

CAGR_15yr = [(PPE(2024) / PPE(2009))^(1/15) - 1] × 100

Calculation Steps:

1. Get total_ppe[2009] and total_ppe[2024] for each district

2. Apply CAGR formula with years = 15

3. For cohort aggregates, use weighted aggregate PPE

Example (Amherst):

- PPE(2009) = $18,234.56

- PPE(2024) = $26,789.45

- CAGR = [(26,789.45 / 18,234.56)^(1/15) - 1] × 100

- CAGR = [1.46923^0.06667 - 1] × 100 = 2.61%

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2. COHORT DETERMINATION CALCULATIONS

Purpose: Assign each district to one of 6 enrollment-based cohorts

Source: school_shared.py, lines 140-200 (calculate_cohort_boundaries)

Cohort Definitions:

- TINY: 0 - Q1 (25th percentile)

- SMALL: Q1 - Q2 (median)

- MEDIUM: Q2 - Q3 (75th percentile)

- LARGE: Q3 - 10,000 FTE

- SPRINGFIELD: > 10,000 FTE (outlier district)

Calculation Steps (FY2024):

1. Get all Western MA traditional districts

2. Get FY2024 FTE enrollment for each district

3. Create array of all enrollments (INCLUDING Springfield):

all_enrollments = [154.7, 287.3, ..., 24,567.8] # 59 districts

4. Calculate percentiles ON FULL DATASET:

- Q1 (25th) = np.percentile(all_enrollments, 25) = 154.7
- Q2 (50th) = np.percentile(all_enrollments, 50) = 520.3
- Q3 (75th) = np.percentile(all_enrollments, 75) = 1,597.6
- P90 (90th) = np.percentile(all_enrollments, 90) = 3,892.1

5. Round to clean boundaries:

- Q1: 154.7 → 200 FTE
- Median: 520.3 → 500 FTE
- Q3: 1,597.6 → 1,600 FTE
- P90: 3,892.1 → 4,000 FTE
- Outlier threshold: Fixed at 10,000 FTE

6. Final FY2024 Cohort Boundaries:

- TINY: 0-200 FTE (13 districts)
- SMALL: 201-500 FTE (16 districts)
- MEDIUM: 501-1,600 FTE (15 districts)
- LARGE: 1,601-10,000 FTE (14 districts)
- SPRINGFIELD: >10,000 FTE (1 district: 24,567.8 FTE)

Note: For historical data, cohorts are calculated year-by-year using that year's enrollment distribution.

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3. WEIGHTED AGGREGATION METHODOLOGY

Purpose: Calculate enrollment-weighted per-pupil expenditure for a group of districts

Source: school_shared.py, lines 1001-1050 (weighted_epp_aggregation)

Formula:

Weighted PPE(category, year) = Σ[PPE(d, cat, yr) × FTE(d, yr)] / Σ[FTE(d, yr)]

where d = district, cat = category, yr = year

Calculation Steps:

1. For each district d in cohort:

- Get PPE by category and year: PPE(d, category, year)
- Get FTE enrollment by year: FTE(d, year)

2. For each category and year:

- Calculate weighted numerator:

numerator(cat, yr) = Σ[PPE(d, cat, yr) × FTE(d, yr)] for all d in cohort

- Calculate total FTE:

denominator(yr) = Σ[FTE(d, yr)] for all d in cohort

- Weighted PPE(cat, yr) = numerator(cat, yr) / denominator(yr)

Example (Medium Cohort, Instruction category, 2024):

Districts in Medium cohort (FY2024): 15 districts

District A: PPE_instruction = $12,345, FTE = 678

District B: PPE_instruction = $13,456, FTE = 891

...

District O: PPE_instruction = $11,234, FTE = 1,234

Numerator = (12,345 × 678) + (13,456 × 891) + ... + (11,234 × 1,234)

= 8,369,910 + 11,989,296 + ... + 13,862,756

= 156,432,890

Denominator = 678 + 891 + ... + 1,234 = 13,456 FTE

Weighted PPE_instruction = 156,432,890 / 13,456 = $11,627.45/pupil

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4. SHADING THRESHOLD CALCULATIONS

Purpose: Determine 5% / 0.5pp thresholds for gradient shading in comparison tables

Source: Appendix A (Threshold Analysis), compose_pdf.py lines 1212-1349

Statistical Analysis (All 59 Western MA Districts):

1. Calculate variation in PPE:

- Mean PPE = $24,237

- Std Dev PPE = $5,462

- Coefficient of Variation (CV) = 5,462 / 24,237 = 0.225 (22.5%)

2. Calculate variation in CAGR:

- Mean CAGR = 6.00 percentage points

- Std Dev CAGR = 3.24 percentage points

- Coefficient of Variation (CV) = 3.24 / 6.00 = 0.540 (54.0%)

3. Key insight: CAGR varies 2.4× more than PPE relative to means (54.0% / 22.5% = 2.4)

4. Evaluate threshold scenarios:

Scenario: 5% PPE / 0.5pp CAGR (SELECTED)

- PPE: 5% = 0.05 × 24,237 = $1,212

- PPE Standard Deviations: 1,212 / 5,462 = 0.22 SD

- CAGR Standard Deviations: 0.50 / 3.24 = 0.15 SD

- Balance ratio: 0.15 / 0.22 = 0.7× (CAGR more sensitive for compound growth)

- PPE flagging rate: ~82% of comparisons

- CAGR flagging rate: ~88% of comparisons

Gradient Shading Bins:

- CAGR bins (absolute percentage points): [0.5pp, 1pp, 1.5pp, 2pp+]

- Dollar bins (relative percent): [5%, 10%, 15%, 20%+]

- Intensity increases with bin: lightest shade → darkest shade

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5. DISTRICT COMPARISON TABLE CALCULATIONS

Purpose: Compare district PPE to baseline (cohort or Western MA aggregate)

Source: compose_pdf.py, lines 425-583 (_build_category_table)

For Each Category (e.g., Instruction, Student Support, etc.):

1. Get latest year (2024) and 5-year-ago baseline (2019)

2. Calculate CAGR:

District CAGR = [(PPE_2024 / PPE_2019)^(1/5) - 1] × 100

Baseline CAGR = [(Baseline_PPE_2024 / Baseline_PPE_2019)^(1/5) - 1] × 100

3. Calculate CAGR difference (absolute percentage points):

CAGR_diff = |District_CAGR - Baseline_CAGR|

4. Determine CAGR shading:

if CAGR_diff < 1.0pp: No shading (white)

elif CAGR_diff < 2.0pp: Lightest shade

elif CAGR_diff < 3.0pp: Light shade

elif CAGR_diff < 4.0pp: Medium shade

else: Darkest shade

Color: Amber if above baseline, Teal if below

5. Calculate dollar difference (relative percent):

Dollar_diff = |PPE_2024 - Baseline_PPE_2024| / Baseline_PPE_2024

6. Determine dollar shading:

if Dollar_diff < 5%: No shading

elif Dollar_diff < 10%: Lightest shade

elif Dollar_diff < 15%: Light shade

elif Dollar_diff < 20%: Medium shade

else: Darkest shade

Color: Amber if above baseline, Teal if below

Example (Amherst Instruction vs Medium Cohort):

- Amherst PPE_2019 = $10,234, PPE_2024 = $12,456

- Medium PPE_2019 = $9,876, PPE_2024 = $11,234

CAGR calculations:

- Amherst CAGR = [(12,456 / 10,234)^0.2 - 1] × 100 = 4.01%

- Medium CAGR = [(11,234 / 9,876)^0.2 - 1] × 100 = 2.60%

- CAGR_diff = |4.01 - 2.60| = 1.41pp → Medium-dark amber shade (in [1pp, 1.5pp) bin)

Dollar calculations:

- Dollar_diff = |12,456 - 11,234| / 11,234 = 10.88%

- 10.88% is in [10%, 15%) range → Light amber shade

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6. ENROLLMENT FTE CALCULATIONS

Purpose: Calculate total FTE enrollment from component categories

Source: school_shared.py, prepare_district_epp_lines()

FTE Categories:

- Foundation Enrollment (low-income, ELL, special ed weighted)

- PK (Pre-Kindergarten)

- K-6 (Kindergarten through 6th grade)

- 7-12 (7th through 12th grade)

Calculation:

Total FTE(year) = Foundation(year) + PK(year) + K6(year) + 712(year)

Example (Amherst 2024):

- Foundation = 1,234.5 FTE

- PK = 89.2 FTE

- K-6 = 678.3 FTE

- 7-12 = 543.1 FTE

- Total FTE = 1,234.5 + 89.2 + 678.3 + 543.1 = 2,545.1 FTE

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7. NSS/CH70 FUNDING CALCULATIONS

Purpose: Calculate Chapter 70 aid and Net School Spending from components

Source: school_shared.py, prepare_district_nss_ch70()

Components:

- Foundation Budget: State-determined minimum spending level

- Required Local Contribution: Municipality's required contribution

- Chapter 70 Aid: State aid = Foundation - Required Local Contribution

- Actual Net School Spending (NSS): Actual spending by district

Calculations:

Ch70 Aid = Foundation Budget - Required Local Contribution

NSS Gap = Actual NSS - Foundation Budget

Total Spending = Required Local Contribution + Ch70 Aid + NSS Gap

Example (Amherst 2024):

- Foundation Budget = $45,678,901

- Required Local Contribution = $38,234,567

- Ch70 Aid = $45,678,901 - $38,234,567 = $7,444,334

- Actual NSS = $52,123,456

- NSS Gap = $52,123,456 - $45,678,901 = $6,444,555

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NOTES

• All dollar values shown are illustrative examples for demonstration purposes

• Actual data values are stored in Appendix C (Data Tables)

• All calculations use enrollment-weighted aggregations for cohort comparisons

• CAGR calculations exclude negative or zero values

• Shading applies independently to CAGR and dollar comparisons

• Year-specific cohort assignments are used for all historical maps and scatterplots

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DETAILED WORKED EXAMPLES

The following sections provide complete worked examples for specific districts and cohorts:

This appendix provides calculation examples showing the source of numbers and demonstrating every step of the analysis for two examples:

1. Western MA Medium Cohort (aggregate cohort calculations)

2. Amherst-Pelham Regional (individual district calculations)

The following pages detail:

• Data sources and cohort determination methodology

• Weighted average per-pupil expenditure calculations

• Time series analysis and compound annual growth rate (CAGR) formulas

• Chapter 70 state aid and Net School Spending calculations

• Comparative analysis methods

PART 1: DATA SOURCES AND COHORT DETERMINATION

All calculations use data from the file "E2C_Hub_MA_DESE_Data.xlsx" which contains multiple sheets:

• Sheet "District Expend by Category": Contains expenditure and enrollment data with columns DIST_NAME, YEAR, IND_CAT, IND_SUBCAT, IND_VALUE, SOURCE

• Sheet "District Regions": Contains regional classifications and school types

• Sheet "profile_DataC70": Contains Chapter 70 and Net School Spending data

Data sources by sheet (per SOURCE column):

District Expend by Category - Expenditures per Pupil:

Source: Massachusetts Department of Elementary and Secondary Education (DESE)

Dataset: "District Expenditures by Spending Category"

Last updated: August 12, 2025

URL: https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/

This dataset provides:

• Expenditures per pupil (EPP) by spending category for all Massachusetts school districts

• Annual data from FY1993 to present

• Eight spending categories: Teachers, Insurance/Retirement/Other, Pupil Services, Other Teaching Services, Operations and Maintenance, Instructional Leadership, Administration, Other

District Expend by Category - Student Enrollment:

Source: Massachusetts Department of Elementary and Secondary Education (DESE)

Dataset: "District Expenditures by Spending Category"

URL: https://educationtocareer.data.mass.gov/Finance-and-Budget/District-Expenditures-by-Spending-Category/er3w-dyti/

This dataset provides three enrollment indicators:

• In-District FTE Pupils: Students enrolled in district-operated schools (used for cohort calculations)

• Out-of-District FTE Pupils: District students placed in other schools

• Total FTE Pupils: Sum of In-District and Out-of-District FTE

profile_DataC70 - Chapter 70 and Net School Spending:

Source: Massachusetts Department of Elementary and Secondary Education (DESE)

Dataset: Chapter 70 District Profiles (compiled from various DESE sources)

URL: Various sources compiled in profile_DataC70 spreadsheet

This dataset provides:

• Chapter 70 Aid (c70aid): State aid received by each district

• Required Net School Spending (rqdnss2): Minimum spending required by state law

• Actual Net School Spending (actualNSS): Total district spending on education

• Foundation Enrollment (distfoundenro): District foundation enrollment used for Ch70 funding calculations

District Regions - Regional Classifications:

Source: DESE district profiles and regional service mappings

This dataset provides:

• EOHHS regional designations (Western MA, Central MA, etc.)

• School type classifications (Traditional, Regional, Charter, Virtual, etc.)

1.1 COHORT BOUNDARY DETERMINATION

Cohort boundaries are determined by reference to Interquartile Range (IQR) analysis on Fiscal Year 2024 Full-Time Equivalent (FTE) in-district enrollment for all traditional public school districts in Western Massachusetts. Cohort boundaries are rounded for easier readability.

Step 1: Extract FY24 In-district FTE enrollment data

From Sheet "District Expend by Category", filter rows where:

• IND_CAT = "Student Enrollment"

• IND_SUBCAT = "In-District FTE Pupils" (Full-Time Equivalent students enrolled in district-operated schools)

• YEAR = 2024

Step 2: Calculate quartile statistics

Using the 60 traditional districts in Western MA with valid data (including Springfield), calculate:

• First quartile (Q1) = 25th percentile = 154.7 In-district FTE

• Second quartile/Median (Q2) = 50th percentile = 792.0 In-district FTE

• Third quartile (Q3) = 75th percentile = 1,597.6 In-district FTE

• 90th percentile (P90) = 3,571.1 In-district FTE

Step 3: Define cohort boundaries

• TINY: 0 to 200 In-district FTE (from zero to Q1 rounded to nearest 100)

• SMALL: 201 to 800 In-district FTE (from max TINY +1 to Median rounded to nearest 100)

• MEDIUM: 801 to 1,600 In-district FTE (from max SMALL +1 to Q3 rounded to nearest 100)

• LARGE: 1,601 to 4,000 In-district FTE (from max MEDIUM +1 to P90 rounded to nearest 1,000)

• X-LARGE: 4,001 to 10K In-district FTE (from max LARGE +1 to fixed threshold of 10,000)

• OUTLIERS (Springfield): Greater than fixed threshold of 10,000 In-district FTE

Step 4: Assign each district to its cohort

For each district, compare its FY24 In-district FTE enrollment to the boundaries above.

Example: Amherst-Pelham Regional has 1,209.3 In-district FTE in FY24

Since 801 ≤ 1,209.3 ≤ 1,600, Amherst-Pelham is assigned to the MEDIUM cohort.

Example: Gill-Montague Regional has 899.8 In-district FTE in FY24

Since 801 ≤ 899.8 ≤ 1,600, Gill-Montague is assigned to the MEDIUM cohort.

Example: Belchertown has 2,067.8 In-district FTE in FY24

Since 1,601 ≤ 2,067.8 ≤ 4,000, Belchertown is assigned to the LARGE cohort.

1.2 EXPENDITURE CATEGORIES

All Per-Pupil Expenditure (PPE) calculations use eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

These categories are derived from the Massachusetts Department of Elementary and Secondary Education (DESE) function code groupings in end-of-year reporting, and provide a standardized breakdown of all district expenditures. Note: District budgeting practices typically use categories that do not match the DESE end-of-year reporting standard.

PART 2: WESTERN MA MEDIUM COHORT CALCULATIONS

The Medium cohort contains 15 districts with in-district enrollment between 801 and 1,600 FTE in FY24. This section demonstrates how weighted average Per-Pupil Expenditure (PPE) is calculated for the cohort.

2.1 MEMBER DISTRICTS

Districts in the Medium cohort (sorted by FY24 In-district FTE enrollment):

1. Mohawk Trail: 804.8 In-district FTE
2. Monson: 807.8 In-district FTE
3. Gill-Montague: 899.8 In-district FTE
4. Amherst: 1,001.5 In-district FTE
5. Hoosac Valley Regional: 1,002.7 In-district FTE

6. Palmer: 1,061.5 In-district FTE
7. Ware: 1,095.8 In-district FTE
8. Berkshire Hills: 1,174.4 In-district FTE
9. North Adams: 1,196.0 In-district FTE
10. Mount Greylock: 1,206.9 In-district FTE
11. Amherst-Pelham: 1,209.3 In-district FTE
12. Southwick-Tolland-Granville Regional School District: 1,312.8 In-district FTE
13. Greenfield: 1,394.0 In-district FTE
14. Easthampton: 1,403.7 In-district FTE
15. Central Berkshire: 1,585.9 In-district FTE

Total In-district FTE for Medium cohort = Sum of all member In-district FTE = 17,156.9 FTE

2.2 WEIGHTED AVERAGE PPE CALCULATION (EXAMPLE: FY2024 TEACHERS)

For each expenditure category, we calculate an enrollment-weighted average across all districts in the cohort.

Step 1: Extract per-pupil expenditure data for Teachers category

From Sheet "District Expend by Category", filter rows where:

- IND_CAT = "Expenditures per Pupil"
- IND_SUBCAT = "Teachers"
- YEAR = 2024

Step 2: For each district, multiply PPE by enrollment weight

Example calculations for first three districts:

Mohawk Trail:

- FY24 Teachers PPE = [actual value from data] per pupil
- FY24 In-district FTE Enrollment = 804.8 FTE
- Weight = 804.8 / 17,156.9 = 0.0469 (4.69% of cohort enrollment)
- Weighted contribution = [Teachers PPE] × 0.0469

Monson:

- FY24 Teachers PPE = [actual value from data] per pupil
- FY24 In-district FTE Enrollment = 807.8 FTE
- Weight = 807.8 / 17,156.9 = 0.0471 (4.71% of cohort enrollment)
- Weighted contribution = [Teachers PPE] × 0.0471

Gill-Montague:

- FY24 Teachers PPE = [actual value from data] per pupil
- FY24 In-district FTE Enrollment = 899.8 FTE
- Weight = 899.8 / 17,156.9 = 0.0524 (5.24% of cohort enrollment)
- Weighted contribution = [Teachers PPE] × 0.0524

[Continue for all 15 districts...]

Step 3: Sum all weighted contributions

Weighted Average Teachers PPE = Sum of all weighted contributions = [calculated value] per pupil

This weighted average calculation is repeated for all eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

Step 4: Calculate total weighted average PPE

Total Weighted Average PPE (FY24) = Sum of all eight category weighted averages

= Teachers + Insurance + Pupil Services + Other Teaching + Operations + Instructional Leadership + Administration + Other

= [sum of actual calculated values] per pupil

2.3 TIME SERIES CALCULATIONS

The same weighted average calculation is performed for each fiscal year from 2009 to 2024, producing a time series of cohort-level PPE values.

For each year Y from 2009 to 2024:

1. Extract FY Y enrollment for all 13 Medium cohort districts
2. Calculate cohort total enrollment for year Y

3. For each of the 8 expenditure categories:

a. Extract PPE values for year Y

b. Calculate enrollment weight for each district in year Y

c. Calculate weighted average PPE for that category in year Y

4. Sum the 8 category weighted averages to get total PPE for year Y

This produces arrays:

• Years: [2009, 2010, 2011, ..., 2024]

• Total PPE: [PPE_2009, PPE_2010, PPE_2011, ..., PPE_2024]

• Teachers PPE: [Teachers_2009, Teachers_2010, ..., Teachers_2024]

• [Similar arrays for other 7 categories]

2.4 COMPOUND ANNUAL GROWTH RATE (CAGR)

CAGR measures the annualized growth rate over multiple years.

Formula:

CAGR = ((Value_end / Value_start)^(1/number_of_years) - 1) × 100%

Where:

• Value_end = PPE in final year (2024)

• Value_start = PPE in initial year (2009)

• number_of_years = 2024 - 2009 = 15 years

Example for Medium cohort Total PPE:

• PPE_2009 = [calculated weighted average for 2009] per pupil

• PPE_2024 = [calculated weighted average for 2024] per pupil

• Number of years = 15

CAGR = ((PPE_2024 / PPE_2009)^(1/15) - 1) × 100%

= [calculated value] % per year

This calculation is performed for:

• Total PPE CAGR

• Each of the 8 category-specific PPE CAGRs

• Chapter 70 state aid CAGR

• Net School Spending (NSS) CAGR

• Enrollment CAGR

2.5 YEAR-OVER-YEAR (YOY) GROWTH RATE

Year-over-Year (YoY) growth rate measures the percentage change from one year to the next, providing insight into annual fluctuations in expenditures.

Formula:

YoY Growth Rate = ((Value_year / Value_previous_year) - 1) × 100%

Where:

• Value_year = PPE in current year

• Value_previous_year = PPE in previous year

Example for Medium cohort Total PPE:

Assume the time series shows:

• PPE_2009 = $12,000 per pupil

• PPE_2010 = $12,500 per pupil

• PPE_2011 = $13,000 per pupil

• PPE_2012 = $13,100 per pupil

• ...

YoY calculations:

• YoY_2010 = (($12,500 / $12,000) - 1) × 100% = 4.17%

• YoY_2011 = (($13,000 / $12,500) - 1) × 100% = 4.00%

• YoY_2012 = (($13,100 / $13,000) - 1) × 100% = 0.77%

• ...

Key differences from CAGR:

• YoY shows year-to-year changes; CAGR smooths growth over multiple years

• YoY captures annual volatility; CAGR assumes constant growth

• YoY is useful for identifying specific years with unusual growth patterns

Example comparison:

If CAGR from 2009 to 2024 is 3.71% per year, but YoY rates vary from 0.5% to 8.0% across individual years, this reveals that growth was not constant—some years had much higher or lower increases than the average.

Use cases:

• Identifying years with budget constraints (low YoY)

• Identifying years with significant increases (high YoY)

• Understanding the impact of specific events (e.g., pandemic, policy changes) on year-to-year spending

This YoY calculation is performed for:

• Total PPE YoY for each year from 2010 to 2024 (15 values)

• Each of the 8 category-specific PPE YoY rates

• Chapter 70 state aid YoY

• Net School Spending (NSS) YoY

• Enrollment YoY

Note: YoY for year N requires data from year N-1, so the first YoY value is for 2010 (comparing 2010 to 2009).

PART 3: AMHERST-PELHAM REGIONAL DISTRICT CALCULATIONS

This section demonstrates all calculations for a single district: Amherst-Pelham Regional.

3.1 DISTRICT PROFILE

District: Amherst-Pelham Regional

Cohort Assignment: MEDIUM (801-1,600 In-district FTE)

FY24 In-district FTE Enrollment: 1,209.3 FTE

Data Source: Appendix A Data Tables

• Enrollment data: See "In-District FTE" column in Appendix A

• PPE data: See expenditure category columns in Appendix A

• Chapter 70 data: See "Chapter 70 Aid" column in Appendix A

• NSS data: See "Required NSS" and "Actual NSS" columns in Appendix A

3.2 ENROLLMENT TIME SERIES

Step 1: Extract in-district enrollment data

From Appendix A Data Tables, locate:

• District: "Amherst-Pelham Regional"

• Find the "In-District FTE" column

• Read values for all years from 2009 to 2024

Step 2: Create time series array

Years: [2009, 2010, 2011, ..., 2023, 2024]

In-district FTE: [FTE_2009, FTE_2010, FTE_2011, ..., FTE_2023, 1,209.3]

(Extract actual values for each year from Appendix A)

Step 3: Calculate enrollment CAGR

CAGR = ((FTE_2024 / FTE_2009)^(1/15) - 1) × 100%

= ((1,209.3 / FTE_2009)^(1/15) - 1) × 100%

= [calculated value] % per year

Where:

• FTE_2024 = 1,209.3 In-district FTE (from data)

• FTE_2009 = [actual value from data] In-district FTE

• Number of years = 15 (2024 - 2009)

A negative CAGR indicates declining enrollment, positive indicates growth.

3.3 PER-PUPIL EXPENDITURE CALCULATIONS

For Amherst-Pelham Regional, PPE is calculated for each of the eight categories:

1. Teachers
2. Insurance, Retirement and Other
3. Pupil Services
4. Other Teaching Services
5. Operations and Maintenance
6. Instructional Leadership
7. Administration
8. Other

Step 1: Extract PPE data for each category

From Appendix A Data Tables, locate:

• District: "Amherst-Pelham Regional"

• Find the columns for each of the 8 expenditure categories

• Read values for all years from 2009 to 2024

Step 2: Example calculation for Teachers category (FY2024)

From Appendix A Data Tables:

• District: "Amherst-Pelham Regional"

• Year: 2024

• Column: "Teachers"

• Value: [actual value from Appendix A] per pupil

This value represents the amount Amherst-Pelham Regional spent per Full-Time Equivalent (FTE) student on Teachers in FY2024.

Step 3: Calculate total PPE for FY2024

Total PPE (FY2024) = Teachers + Insurance, Retirement and Other + Pupil Services + Other Teaching Services + Operations and Maintenance + Instructional Leadership + Administration + Other

Using column IND_VALUE for YEAR = 2024 and each category:

• Teachers: [actual value]

• Insurance, Retirement and Other: [actual value]

• Pupil Services: [actual value]

• Other Teaching Services: [actual value]

• Operations and Maintenance: [actual value]

• Instructional Leadership: [actual value]

• Administration: [actual value]

• Other: [actual value]

Total PPE (FY2024) = [sum of actual values] per pupil

Step 4: Repeat for all years 2009-2024

For each year from 2009 to 2024, extract IND_VALUE for each of the 8 categories and sum them to get total PPE for that year.

This produces time series arrays:

- Years: [2009, 2010, ..., 2024]
- Total PPE: [PPE_2009, PPE_2010, ..., PPE_2024]
- Teachers: [Teachers_2009, Teachers_2010, ..., Teachers_2024]
- Insurance, Retirement and Other: [Insurance_2009, Insurance_2010, ..., Insurance_2024]
- Pupil Services: [Pupil_Services_2009, Pupil_Services_2010, ..., Pupil_Services_2024]
- Other Teaching Services: [Other_Teaching_2009, Other_Teaching_2010, ..., Other_Teaching_2024]
- Operations and Maintenance: [Operations_2009, Operations_2010, ..., Operations_2024]
- Instructional Leadership: [Instructional_Leadership_2009, Instructional_Leadership_2010, ..., Instructional_Leadership_2024]
- Administration: [Administration_2009, Administration_2010, ..., Administration_2024]
- Other: [Other_2009, Other_2010, ..., Other_2024]

Step 5: Calculate CAGR for each category

Formula: CAGR = ((Value_2024 / Value_2009)^(1/15) - 1) × 100%

Example for Teachers:

- Teachers_2009 = [actual value from data] per pupil
- Teachers_2024 = [actual value from data] per pupil
- CAGR_Teachers = ((Teachers_2024 / Teachers_2009)^(1/15) - 1) × 100%

= [calculated value] % per year

Repeat this calculation for all 8 categories plus total PPE to get 9 CAGR values.

3.4 CHAPTER 70 STATE AID CALCULATIONS

Chapter 70 is the major state aid program for Massachusetts K-12 public schools.

Step 1: Extract Chapter 70 data

From Appendix A Data Tables, locate:

- District: "Amherst-Pelham Regional"
- Column: "Chapter 70 Aid"
- Read values for all years from 2009 to 2024

Step 2: Extract total district expenditures

From Appendix A Data Tables, calculate total expenditure as:

- District: "Amherst-Pelham Regional"
- Sum of all expenditure categories for each year
- Alternatively, use "Actual NSS" column if available

Step 3: Calculate Chapter 70 as percentage of total expenditure (FY2024 example)

- Chapter 70 Aid (FY2024) = [value from "Chapter 70 Aid" column in Appendix A for 2024]
- Total Expenditure (FY2024) = [sum of expenditure categories from Appendix A for 2024]
- Percentage = (Chapter 70 Aid / Total Expenditure) × 100%

This percentage shows what portion of the district's total expenditure came from Chapter 70 state aid in FY2024.

Step 4: Calculate Chapter 70 per-pupil (FY2024 example)

• Chapter 70 Aid (FY2024) = [value from "Chapter 70 Aid" column in Appendix A]

• Enrollment (FY2024) = 1,209.3 In-district FTE

• Chapter 70 Per-Pupil = Chapter 70 Aid / 1,209.3 FTE = [calculated value] per FTE

Step 5: Calculate Chapter 70 CAGR

• Chapter 70 Aid (2009) = [value from "Chapter 70 Aid" column in Appendix A for 2009]

• Chapter 70 Aid (2024) = [value from "Chapter 70 Aid" column in Appendix A for 2024]

• CAGR = ((Ch70Aid_2024 / Ch70Aid_2009)^(1/15) - 1) × 100%

= [calculated value] % per year

A negative CAGR indicates that Chapter 70 aid decreased, positive indicates growth.

3.5 NET SCHOOL SPENDING (NSS) CALCULATIONS

Net School Spending (NSS) is the total amount a school district spends on education from all sources, excluding certain items like transportation, debt service, and capital expenditures.

Step 1: Extract NSS data

From Appendix A Data Tables, locate:

• District: "Amherst-Pelham Regional"

• Columns: "Required NSS" and "Actual NSS"

• Read values for all years from 2009 to 2024

Step 2: Calculate NSS per-pupil (FY2024 example)

• Total NSS (FY2024) = [value from "Actual NSS" column in Appendix A for 2024]

• Enrollment (FY2024) = 1,209.3 In-district FTE

• NSS Per-Pupil = Total NSS / 1,209.3 FTE = [calculated value] per FTE

Step 3: Calculate NSS CAGR

• NSS (2009) = [value from "Actual NSS" column in Appendix A for 2009]

• NSS (2024) = [value from "Actual NSS" column in Appendix A for 2024]

• CAGR = ((NSS_2024 / NSS_2009)^(1/15) - 1) × 100%

= [calculated value] % per year

This indicates the average annual growth rate of NSS from 2009 to 2024.

Step 4: Compare NSS per-pupil to cohort average

Using the weighted average calculation method described in Part 2, the MEDIUM cohort average NSS per-pupil (FY2024) = [calculated weighted average] per FTE.

Amherst-Pelham's NSS per-pupil ([actual value from calculation]) is [percentage difference]% [above/below] the MEDIUM cohort average.

Calculation: ((Amherst-Pelham NSS per-pupil - MEDIUM cohort average) / MEDIUM cohort average) × 100%

3.6 COMPARATIVE ANALYSIS

All district-specific calculations can be compared to:

1. Cohort averages (in this case, MEDIUM cohort weighted averages for Amherst-Pelham Regional)

2. Regional averages (all Western MA traditional districts)

3. Statewide averages (if available)

For each metric (Total PPE, each of 8 category PPEs, Chapter 70 per-pupil, NSS per-pupil, enrollment growth), calculate:

• District value

• Cohort average

• Difference = District value - Cohort average

• Percentage difference = (Difference / Cohort average) × 100%

These comparisons reveal whether a district is spending more or less than similar-sized districts, and where specific spending differences occur across the eight categories:

1. Teachers

2. Insurance, Retirement and Other

3. Pupil Services

4. Other Teaching Services

5. Operations and Maintenance

6. Instructional Leadership

7. Administration

8. Other

Appendix C. Data Tables

All data values used in plots

This appendix contains the underlying data tables for all districts and regions shown in the report. Each table shows PPE by category (in $/pupil), FTE enrollment counts, and NSS/Ch70 funding components (in absolute dollars) across all available years.

Data: All Western MA Traditional Districts: Tiny (0-200 FTE)

PPE ($/pupil), FTE Enrollment, and NSS/Ch70 Funding ($)

PPE by Category ($/pupil)

Table 1

Category	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Other	$577	$571	$623	$643	$647	$665	$676	$709	$780	$579	$673	$738	$1,380	$1,042	$1,304	$1,057
Administration	$770	$808	$814	$853	$915	$851	$774	$819	$863	$869	$770	$833	$950	$1,003	$1,114	$1,126
Instructional Leadership	$1,042	$1,096	$1,125	$1,225	$1,269	$1,316	$1,496	$1,584	$1,510	$1,524	$1,543	$1,589	$1,817	$1,733	$1,777	$1,991
Operations and Maintenance	$1,374	$1,312	$1,460	$1,416	$1,359	$1,558	$1,554	$1,452	$1,433	$1,468	$1,811	$1,670	$2,060	$2,389	$2,232	$2,652
Other Teaching Services	$1,899	$1,926	$1,987	$2,112	$2,412	$2,378	$2,494	$2,635	$2,673	$2,707	$3,004	$3,115	$3,322	$3,677	$3,638	$3,757
Pupil Services	$1,669	$1,792	$1,802	$1,899	$2,023	$2,061	$2,145	$2,206	$2,290	$2,261	$2,323	$2,204	$2,695	$2,673	$2,954	$3,406
Insurance, Retirement and Other	$2,258	$2,405	$2,595	$2,737	$2,926	$3,148	$3,125	$3,143	$3,905	$3,390	$3,662	$3,769	$4,357	$4,048	$4,506	$5,009
Teachers	$5,148	$5,382	$5,545	$5,277	$5,430	$5,906	$6,187	$6,222	$6,245	$6,520	$6,792	$6,982	$8,015	$7,858	$7,808	$8,507
Total	$14,738	$15,291	$15,950	$16,161	$16,981	$17,882	$18,452	$18,770	$19,700	$19,319	$20,577	$20,900	$24,596	$24,423	$25,334	$27,504

FTE Enrollment

Table 2

FTE Series	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
In-District FTE Pupils	131	127	125	125	123	120	122	116	121	123	122	122	107	115	114	113
Foundation Enrollment	123	120	116	109	109	108	107	111	108	107	105	102	101	91	93	94
Out-of-District FTE Pupils	41	41	41	37	40	38	37	39	38	36	35	33	31	34	34	32

NSS/Ch70 Funding Components ($)

Table 3

Component	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Ch70 Aid	$3K	$3K	$3K	$3K	$3K	$3K	$3K	$4K	$4K	$4K	$4K	$4K	$4K	$4K	$4K	$5K
Req NSS (minus Ch70)	$5K	$6K	$6K	$6K	$7K	$7K	$7K	$7K	$7K	$7K	$7K	$7K	$8K	$7K	$8K	$8K
Actual NSS (minus Req NSS)	$5K	$4K	$5K	$5K	$5K	$6K	$6K	$6K	$8K	$7K	$8K	$8K	$10K	$10K	$10K	$10K
Total NSS	$13K	$14K	$15K	$15K	$15K	$16K	$16K	$17K	$18K	$18K	$19K	$19K	$22K	$21K	$22K	$23K

Appendix D. Additional Visualizations

Scatterplot of enrollment vs. per-pupil expenditure with quartile boundaries (2019)

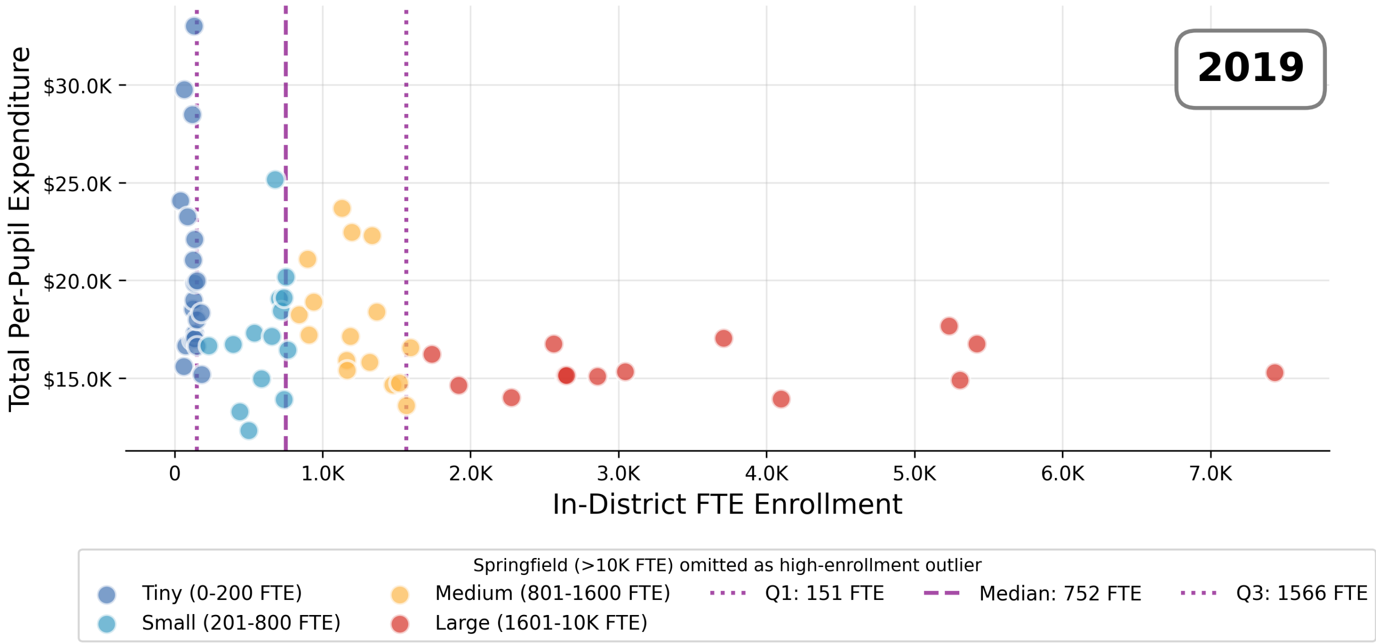

Figure 1

This appendix contains historical scatterplots and geographic maps showing the evolution of district enrollment and per-pupil expenditures over time. These visualizations provide context for understanding how enrollment patterns and spending levels have shifted across Western Massachusetts districts from 2009 to 2024.

This scatterplot shows the relationship between district enrollment (x-axis) and per-pupil expenditure (y-axis) for Western Massachusetts traditional districts in 2019. Each point represents one district, colored by enrollment cohort. Horizontal lines show quartile boundaries used to define cohorts.